PROBLEM 38

Toroidal Geometry

Particle on Torus

Original statement · Problems.pdf, p. 26

A heavy point particle P with mass $m_P = 1$ moves without friction on a vertically-standing torus. The particle cannot separate from the surface of this torus. The torus is defined in the Cartesian coordinates of the inertial reference frame S_0 by the following expressions:

$$\begin{cases} x_0 = (2 + \cos\phi)\sin\theta \\ y_0 = \sin\phi \\ z_0 = 4 + (2 + \cos\phi)\cos\theta \end{cases}$$

A second particle Q with mass $m_Q = 2$ is constrained to move on the Ox_0y_0 plane without friction. The two particles P and Q attract each other with a force that is proportional to the distance between them. The proportionality constant of this force is $k = 2$. Initially, the particle P is located at $\mathbf{r}_0^P(t_0) = 2\mathbf{i}_0 + \mathbf{j}_0 + 4\mathbf{k}_0$ with $\mathbf{v}_0^P(t_0) = -\mathbf{k}_0$ and the particle Q is located at $\mathbf{r}_0^Q(t_0) = 3\mathbf{i}_0 + \mathbf{j}_0$ with $\mathbf{v}_0^Q(t_0) = -\mathbf{i}_0 + \mathbf{j}_0$.

(a) Pick a minimal, valid set of coordinates to describe the configuration of the system. Indicate what is the geometrical meaning of the chosen coordinates.

(b) Obtain a general expression for the position, velocity, and acceleration vectors of P and Q in S_0 using the chosen coordinates and their derivatives.

(c) Obtain two independent tangent vectors and one normal vector to the torus as a function of the chosen coordinates.

(d) Identify and express mathematically all the forces acting on P and Q.

(e) Find down a set of second order differential equations to determine the evolution of the chosen coordinates in time, free of any other unknowns.

(f) Obtain a scalar conservation law.

(g) Attending to this conservation law alone, can P ever reach the highest point of the torus?

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:**
- Particle P ($m_P = 1$) moves without friction on the vertical torus:

$$x_0 = (2 + \cos\phi)\sin\theta, \quad y_0 = \sin\phi, \quad z_0 = 4 + (2 + \cos\phi)\cos\theta$$

- Particle Q ($m_Q = 2$) is constrained to the horizontal plane Ox_0y_0 ($z_0 = 0$).
- Mutual attractive spring force: $\mathbf{F}_{PQ} = -k(\mathbf{r}_P - \mathbf{r}_Q)$ with $k = 2$.
- **Initial State (t_0):**
- P: $\mathbf{r}_P(t_0) = 2\mathbf{i}_0 + \mathbf{j}_0 + 4\mathbf{k}_0$, $\mathbf{v}_P(t_0) = -\mathbf{k}_0$.
- Q: $\mathbf{r}_Q(t_0) = 3\mathbf{i}_0 + \mathbf{j}_0$, $\mathbf{v}_Q(t_0) = -\mathbf{i}_0 + \mathbf{j}_0$.
- **Parameters:** $m_P = 1, m_Q = 2, k = 2, g = 9.81$ m/s^2.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Minimal coordinate set:
- For P: Toroidal angle θ and poloidal angle ϕ.
- For Q: Cartesian planar coordinates (ξ, η) in the plane $z = 0$.
- Total CDOFs: $2 + 2 = 4$.
- **Toroidal basis.** With the toroidal angle θ (rotation about $\mathbf{j}_0$), $\mathcal{B}_1 = \{\mathbf{e}_\rho, \mathbf{e}_\theta, \mathbf{j}_0\}$ with $\mathbf{e}_\rho = \sin\theta\,\mathbf{i}_0 + \cos\theta\,\mathbf{k}_0$ and $\mathbf{e}_\theta = \cos\theta\,\mathbf{i}_0 - \sin\theta\,\mathbf{k}_0$ (so that $\mathbf{e}_\rho \times \mathbf{e}_\theta = \mathbf{j}_0$). The columns are the components in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Poloidal basis.** With the poloidal angle ϕ (rotation about $\mathbf{e}_\theta$), $\mathcal{B}_2 = \{\mathbf{n}, \mathbf{e}_\theta, \mathbf{t}_\phi\}$ with $\mathbf{n} = \cos\phi\,\mathbf{e}_\rho + \sin\phi\,\mathbf{j}_0$ (outward normal) and $\mathbf{t}_\phi = -\sin\phi\,\mathbf{e}_\rho + \cos\phi\,\mathbf{j}_0$. Its components in $\mathcal{B}_1$ are the columns of

$$[{}_1R_2] = \begin{pmatrix} \cos\phi & 0 & -\sin\phi \\ 0 & 1 & 0 \\ \sin\phi & 0 & \cos\phi \end{pmatrix}, \qquad \det[{}_1R_2] = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

- In these bases $\mathbf{r}_P = 4\mathbf{k}_0 + 2\mathbf{e}_\rho + \mathbf{n}$, and the two tangent vectors and the normal asked in the statement are $\partial\mathbf{r}_P/\partial\theta = (2 + \cos\phi)\,\mathbf{e}_\theta$, $\partial\mathbf{r}_P/\partial\phi = \mathbf{t}_\phi$ and $(\partial\mathbf{r}_P/\partial\theta) \times (\partial\mathbf{r}_P/\partial\phi) = (2 + \cos\phi)\,\mathbf{n}$. The initial position $\mathbf{r}_P(t_0) = 2\mathbf{i}_0 + \mathbf{j}_0 + 4\mathbf{k}_0$ corresponds to $\theta = \phi = \pi/2$. Both matrices and these vector identities were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

Part (a)-(d): Force on P due to Q

The attractive force exerted by Q on P is:

$$\mathbf{F}_P = k(\mathbf{r}_Q - \mathbf{r}_P) = 2(\mathbf{r}_Q - \mathbf{r}_P)$$

Substitute $\mathbf{r}_Q = \xi\mathbf{i}_0 + \eta\mathbf{j}_0$ and $\mathbf{r}_P$:

$$\mathbf{F}_P = 2[\xi - (2+\cos\phi)\sin\theta]\mathbf{i}_0 + 2[\eta - \sin\phi]\mathbf{j}_0 - 2[4 + (2+\cos\phi)\cos\theta]\mathbf{k}_0$$

Compare with the official solution key (Problems.pdf, Solution keys section).

Part (e)-(g): Scalar Conservation Law & Reachability of Torus Summit

- Both gravity and the mutual attractive spring force are conservative. The normal constraint forces do zero work.
- Total mechanical energy $E = T_P + T_Q + V_g + V_{spring} = \text{constant}$.
- Evaluating total initial energy E_0:

$$T_P(t_0) = \frac{1}{2}(1)(1)^2 = \frac{1}{2}$$

$$T_Q(t_0) = \frac{1}{2}(2)((-1)^2 + 1^2) = 2$$

$$V_g(t_0) = m_P g z_P(t_0) = 1 \cdot g \cdot 4 = 4g$$

$$|\mathbf{r}_P - \mathbf{r}_Q|^2 = (2-3)^2 + (1-1)^2 + (4-0)^2 = 1 + 0 + 16 = 17$$

$$V_{spring}(t_0) = \frac{1}{2}k|\mathbf{r}_P - \mathbf{r}_Q|^2 = \frac{1}{2}(2)(17) = 17$$

$$E_0 = \frac{1}{2} + 2 + 4g + 17 = 19.5 + 4g$$

- At the highest point of the torus, $z_{max} = 4 + (2+1) = 7$.

The minimum gravitational potential alone required to reach this point is:

$$V_{g,max} = m_P g z_{max} = 7g$$

Required energy: $E_{req} \geq 7g$.

Since $19.5 + 4g < 7g \iff 19.5 < 3g \approx 29.4$, the particle **does not possess enough energy** to ever reach the highest point of the torus!

PHASE 4

Physical Interpretation & Dimensional Verification

- Energy budgeting definitively rules out reaching the summit regardless of the complex non-linear 4-DOF coupling.
- **Dimensional check (SI):** $[k\,|\mathbf{r}_P - \mathbf{r}_Q|^2] = \mathrm{N/m}\cdot\mathrm{m}^2 = \mathrm{J}$, $[m\,g\,z] = \mathrm{kg}\cdot\mathrm{m/s^2}\cdot\mathrm{m} = \mathrm{J}$ and $[m\,v^2] = \mathrm{J}$, so every term of E_0 is an energy.
- **Numerical values ($g = 9.81\ \mathrm{m/s^2}$):** $E_0 = 19.5 + 4g = 58.74\ \mathrm{J}$, whereas reaching the summit $z_{max} = 7\ \mathrm{m}$ requires at least $m_P g z_{max} = 7g = 68.67\ \mathrm{J}$ (the kinetic and spring energies are non-negative, so the potential energy at the summit cannot exceed E_0). The deficit is $68.67 - 58.74 = 9.93\ \mathrm{J} > 0$, hence the summit is unreachable.
- **Robustness:** the same inequality holds for any spring constant that keeps $V_{spring} \geq 0$ and also for $k \to 0$ (then $E_0 = 2.5 + 4g = 41.74\ \mathrm{J} < 7g$), so the conclusion does not depend on the value of k.

Source: Problems.pdf (Page 26)